

OTHMANE FERRAH

Data Scientist / AI Engineer — Intern

Master's in Data Science & AI • Machine Learning, Computer Vision, LLMs & VLMs

☎ +212 636 85 13 43 ✉ 0x0red.me@gmail.com in [LinkedIn](#) 📍 Meknès, Morocco

Profile

Master's student in Data Science & AI with a strong mathematical foundation, seeking a data science / AI internship. Hands-on across the ML stack — statistics and time-series, computer vision, NLP, LLMs/VLMs, and graph learning — through applied projects spanning medical imaging (VLMs + LoRA), retrieval-augmented (RAG) chatbots, sensor-data forecasting, and real-time gesture recognition. Comfortable owning a problem end-to-end: data pipeline, model fine-tuning, evaluation, and deployment. Co-founder of two AI startups.

Education

Université Moulay Ismaïl — Faculté des Sciences <i>Master's — Parcours d'Excellence, Data Science & AI</i>	Meknès, Morocco <i>Sep 2025 – Sep 2027</i>
Université Moulay Ismaïl — Faculté des Sciences <i>Licence — Parcours d'Excellence, Data Science & AI</i>	Meknès, Morocco <i>Nov 2024 – Jul 2025</i>
Université Moulay Ismaïl — Faculté des Sciences <i>DEUG — Mathematics & Computer Science</i>	Meknès, Morocco <i>2019 – 2022</i>

Technical Skills

Programming: Python, Java, JavaScript/TypeScript, SQL
Machine Learning & AI: Deep Learning, Computer Vision, NLP, LLMs & VLMs, RAG, Graph Neural Networks, PEFT (LoRA)
Statistics & Modeling: Statistics, Time-Series Forecasting, Predictive Modeling, Data Mining, Data Analysis, Mathematical Modeling
Frameworks & Libraries: PyTorch, TensorFlow/Keras, Scikit-learn, XGBoost, LightGBM, OpenCV, MediaPipe, YOLO, PyTorch Geometric, Next.js, Spring Boot, Django
Data & Tools: PostgreSQL, MySQL, pgvector, Ollama, Git, Jupyter
Design & Creative: UX/UI Design, Data Visualization, Figma, Adobe Creative Suite
Languages: Arabic (Native), French (Full Professional), English (Native/Bilingual), Italian (Elementary)

Projects

Brain Tumor MRI Classification — Deep Learning & Vision-Language Models	May 2026 – Present
Benchmarked handcrafted features (GLCM, LBP, HOG) and 8+ ML models against deep learning and VLMs fine-tuned with LoRA for MRI tumor classification. <i>Python, PyTorch, Scikit-learn, XGBoost, VLMs.</i>	
Real-Estate Recommendation Chatbot (RAG)	Mar 2025 – May 2025
Built a full-stack property platform with a RAG chatbot over Llama 3 (Ollama, Spring AI) and a vector database for grounded, hallucination-free recommendations. <i>Next.js, Spring Boot, PostgreSQL/pgvector.</i>	
Sign Language Recognition System	Apr 2025 – Jul 2025
Real-time gesture and action recognition pipeline combining MediaPipe landmarks, YOLO localization, and LSTM temporal modeling. <i>Python, YOLO, MediaPipe, TensorFlow/Keras.</i>	
Social-Graph Community Detection (Louvain, Leiden & GNN)	May 2026 – Jul 2026
Compared modularity-based algorithms with a Graph Neural Network approach on large social graphs (Pokec, YouTube) to power a community-driven recommender. <i>Python, NetworkX, PyTorch Geometric.</i>	
Asteroid Sprint — Arcade Game with AI Hand-Gesture Controls	Feb 2025 – Jul 2025
Shipped a webcam-controlled arcade game using MediaPipe hand tracking (steer / shield / repulse gestures); released publicly on itch.io under GPL-3.0. <i>Python, pygame-ce, ModernGL, OpenCV.</i>	
Intelligent Agriculture System — Weather Station, ET₀ Prediction & Auto-Irrigation	2025
Built an IoT + ML pipeline that forecasts reference evapotranspiration (ET₀) from time-series weather-station sensor data (Penman-Monteith & ML) to drive automated, water-efficient irrigation. <i>Python, Time-Series, Machine Learning, IoT.</i>	
Self-Balancing Robot — Two-Wheeled Inverted Pendulum with PID Control	Sep 2025 – Dec 2025
Kept a two-wheeled robot upright with a tuned PID controller and gyro/accelerometer sensor fusion (complementary filter) on an MPU6050 IMU. <i>Arduino, C/C++, control systems.</i>	

- Immersive VR experience for exploring planets and orbital mechanics in real time; built with the CS Club and showcased at the Moroccan Gaming Expo. *VR, game development, 3D.*

GestMag — Store Management Application

Mar 2025 – Jul 2025

- Desktop app centralizing inventory, sales, customers, and automated invoicing with a live activity dashboard and role-based access. *WinDev, HFSQL.*

FSUMI-LIB — Library Management Platform (Agile/Scrum)

Apr 2024 – Jun 2024

- Web platform for book search, reservations, and loan tracking with an admin dashboard, delivered iteratively using Scrum. *Django, MySQL, Tailwind CSS.*

Professional Experience

NebrasAI*Co-Founder & Creative Director***Meknès, Morocco***Sep 2024 – Present*

- Co-founded an AI startup; shaped product direction and led the full brand and design identity across products and communication.

Sihati*Data Extraction Developer & Graphic Designer***Fès-Meknès, Morocco***Sep 2025 – Present*

- **Data Extraction:** Built scripts and workflows to extract and structure raw sources into clean, usable datasets for the product.
- **Design & Video:** Produced brand visuals and video content for the health platform.

MowajjihAI*Lead Designer (Co-Founder)***Meknès, Morocco***Jan 2023 – Sep 2024*

- Co-founded an AI-driven student-orientation platform and led its product design and brand identity.

Twareg Esports • RAWD Newton*Graphic Designer***Freelance***2025 – 2026*

- Delivered brand assets, logo, and visual identity work for an esports brand (long-term) and an education client.

Leadership & Community

Computer Science Club — FSM (UMI)*President • Lead Designer • Former Communications Manager***Meknès, Morocco***Jul 2022 – Present*

- **President (since Nov 2024):** Lead the club's strategy, events, and community growth; drove AI/CV project teams to the Moroccan Gaming Expo.
- **Lead Designer (since Jul 2022):** Own the club's visual identity and event branding; previously Communications Manager (2023–2024).

Awards, Certifications & Interests

Awards: Best Young Enterprise 2023 (Meknès region); Best Service Award; Responsible Entrepreneur Award; 2nd Prize — "Out of The Box" competition.

Certifications: Company Program regional competition; "Personal Branding: Steps to Brand Yourself"; Organizing committee — 10th Conference on Environmental Instrumentation and Measurements.

Interests: Photography (16.5M+ views portfolio), Programming, Security, Research, Gaming.